

CHAPTER 5 – Renewable Energy Ordinance Specific Development Standards

PART 1 INTRODUCTION

- 5-1 **AUTHORITY**** The requirements of this Ordinance shall apply to all Renewable Energy Systems proposed after the effective date of this Ordinance. Energy Systems for which a required Dubuque County permit and rezoning to Renewable Energy Overlay Zoning District has been properly issued prior to the effective date of this Ordinance shall not be required to meet the requirements of this Ordinance; provided, that any such pre-existing Renewable Energy System, which does not provide energy for a continuous period of twelve (12) months, shall meet the requirements of this Ordinance prior to recommencing production of energy. Also, no modification or alteration to an existing Renewable Energy System shall be allowed without an approved amendment from the Board of Supervisors.
- 5-2 **DEFINITIONS**** For use in this Ordinance, the following terms or words shall be interpreted or defined as follows:
- 5-2.1 “Accessory Solar Energy Conversion System (ASECS)”** A solar energy conversion system which is incidental and subordinate to a principal use on the same parcel and intended to primarily provide electrical power for use on the site in which the system is located.
- 5-2.2 “Accessory Wind Energy Conversion System (AWECS)”** A wind energy conversion system that is incidental and subordinate to a principal use on the same parcel and intended to primarily provide power for use on the site in which the system is located.
- 5-2.3 “Authorized Factory Representative”** An individual with technical training on USWECS, USSECS, USBESS who has received factory installation instructions and is certified in writing by the manufacturer of the system.
- 5-2.4 “Utility Scale Battery Energy Storage Systems (USBESS)”** A bank of batteries or capacitors used to store electricity for later use primarily off-site through the electrical grid or export to the wholesale market – sometimes called a battery storage power station. This definition does not include a battery storage system which is incidental and subordinate to a principal use on the

same parcel and intended to primarily provide electrical power for use on the site in which the system is located.

- 5-2.5 “Building-Integrated Solar Energy Conversion Systems”** A solar energy conversion system (SECS) that is an integral part of a principal or accessory building, rather than a separate mechanical device, replacing or substituting for an architectural or structural component of the building. Building integrated systems include but are not limited to photovoltaic or hot water solar energy systems that are contained within roofing materials, windows, skylights, and awnings.
- 5-2.6 “Building Mounted Wind Energy Conversion System”** A WECS which is securely fastened to any portion of a principal building in order to achieve desired elevation, whether attached directly to the principal building or attached to a tower structure which is in turn fastened to the principal building. These systems are prohibited by this section.
- 5-2.7 “Concentrating Solar Power System (CSPS)”** A power generation system that uses mirrors to concentrate the sun's energy to drive traditional steam turbines or engines that create electricity. The thermal energy concentrated in a CSPS plant is sometimes stored and later used to produce electricity.
- 5-2.8 “Distribution Substation”** An electric system substation that takes high transmitted voltages and reduces the voltage for further distribution to end users with output voltages generally below 35 kV.
- 5-2.8 “Free Standing Wind Energy Conversion System”** A WECS which is elevated by means of a monopole tower only and is not located on another supporting structure except that the tower shall have an appropriately constructed concrete base. Guy-wire supported mast, lattice, or other non-monopole style towers shall not meet this definition.
- 5-2.9 “Ground-Mounted Solar Energy Conversion System”** A SECS mounted on a rack or pole that rests or is attached to the ground.
- 5-2.10 “Height, Total Wind Energy Conversion System”** The height above grade of the system, including the generating unit and the highest vertical extension of any blades or rotors.
- 5-2.11 “Horizontal Axis Wind Energy Conversion System”** A WECS that has blades which rotate through a horizontal plane.

- 5-2.12 “Manual and Automatic Controls for a Wind Energy Conversion System”** Controls that give protection to power grids and limit rotation of the blades to below the designed limits of the WECS.
- 5-2.13 “Maximum Tilt”** The maximum angle of a solar panel or collector (i.e., most vertical position) for capturing solar radiation as compared to the horizon line.
- 5-2.14 “Minimum Tilt”** The minimal angle of a solar panel or collector (i.e., most horizontal position) for capturing solar radiation as compared to the horizon line.
- 5-2.15 “Non-Participating Lot(s)”** One or more lots for which there is not a signed lease or easement for development of a principal-use USSECS, USWECS, or USBESS associated with the applicant project.
- 5-2.16 “Off Grid”** An electrical system that is not connected to utility distribution and transmission facilities or to any building or structure that is connected.
- 5-2.17 “Operator”** The entity or individual that operates a WECS, SECS, or BESS.
- 5-2.18 “Owner”** The entity or individual that has ownership over a WECS, SECS, or BESS.
- 5-2.19 “Participating Lot(s)”** One or more lots under a signed lease or easement for development of a ~~utility-scale~~US-WECS, ~~utility-scale~~US-SECS, or USBESS associated with the applicant project.
- 5-2.20 “Professional Engineer”** An engineer licensed to practice in the State of Iowa.
- 5-2.21 “Prototype and/or Experimental WECS”** An original or homemade, first full scale, new type of functional model.
- 5-2.22 “Roof-Mounted Solar Energy Conversion System”** A SECS mounted on a rack that is fastened to or ballasted on a structure roof.
- 5-2.23 “Rotor”** The portion of the WECS, typically consisting of blades, shafts and hubs that rotate in response to wind and convert the motion into electrical energy. Rotors are often horizontal or vertical.

- 5-2.24 “Shadow Flicker”** The changing of light intensity caused by sunlight through the moving blades of a WECS.
- 5-2.25 “Solar Access”** Unobstructed access to direct sunlight on a lot or building through the entire year, including access across adjacent parcel air rights, for the purpose of capturing direct sunlight to operate a solar energy conversion system.
- 5-2.26 “Solar Carport”** A SECS of any size that is installed on a carport structure that is accessory to a parking area, and which may include electric vehicle supply equipment or energy storage facilities.
- 5-2.27 “Solar Energy Conversion System (SECS)”** A photovoltaic system or solar thermal system for generating electricity or heat, including all above and below ground equipment or components required for the system to operate properly and to be secured to a building surface or the ground. This includes any necessary operations and maintenance building(s), but does not include any temporary construction offices, substation(s), battery energy storage systems (BESS), or transmission facilities between the SECS and the point of interconnection to the electric grid. This definition does not include any concentrating solar power systems (CSPS).
- 5-2.28 “Solar Mounting Devices”** Racking, frames, or other devices that allow the mounting of a solar panel or solar collector onto a roof surface or the ground.
- 5-2.29 “Tower for a Wind Energy Conversion System”** The vertical component of a WECS that elevates the wind turbine generator and attached blades above the ground.
- 5-2.29 “Transmission Substation”** An electric system substation that operates at higher voltages usually above 110 kV for long-distance power transmission.
- 5-2.30 “Utility Scale Wind Energy Conversion System (USWECS)”** One or more wind energy conversion systems located together and used for production of electric power to be interconnected into the local utility electrical grid and primarily for consumption by on-grid utility customers located off the property. USWECS are typically the principal use of the parcel on which located but may include a dual use such as vegetable farming. Commonly referred to as a “wind farm.”

- 5-2.31 “Utility Scale Solar Energy Conversion System (USSECS)”** A solar energy conversion system that is used to produce electrical power for the primary purpose of off-site use through the electrical grid or export to the wholesale market. USSECS do not include concentrating solar power systems (CSPS).
- 5-2.32 “Vertical Axis Wind Energy Conversion System”** A WECS that has blades which rotate through a vertical plane.
- 5-2.33 “Wind Energy Conversion System (WECS)”** Any device, such as a wind charger or wind turbine, which converts wind to a form of usable electric energy. d
- 5-3 PURPOSE** To promote the effective and efficient use of wind energy conversion systems (USWECS), ~~and~~ solar energy conversion systems (USSECS), and battery energy storage systems (USBESS) by regulating and requiring a rezoning for a Renewable Energy Overlay Zoning District for the siting, design, and installation of conversion systems to protect the public health, safety, and welfare of present and future residents of Dubuque County. In addition, this ordinance provides a permitting process for these energy systems to ensure compliance with the provisions of the requirements and standards established or referenced herein. The provisions of this ordinance shall not guarantee wind or solar rights or establish access to the wind or sunlight.
- 5-4 APPLICABILITY** It shall be unlawful to construct, erect, install, alter, or locate any Utility Scale Wind Energy Conversion System (USWECS) Utility Scale Solar Energy Conversion System (USSECS) and Utility Scale Battery Energy Storage System (USBESS) within unincorporated Dubuque County, without first obtaining approval of rezoning to Renewable Energy Overlay Zoning District from Dubuque County as outlined in this ordinance and in compliance with the provisions of this chapter and the Dubuque County Zoning Ordinance. No permit application for a USWECS, USSECS, or USBESS permit shall be granted without first submitting all required information and obtaining necessary rezoning, permits, certifications, and documentation.

PART 2

ZONING AND PERMITS

5-5.5 Application and Procedures

- A. Accessory Wind Energy Conversion Systems and Accessory Solar Energy Conversion Systems.
 - 1. Permit Application Requirements. Application for approval of a permit to construct an AWECS or ASECS shall be submitted to the Zoning Administrator on a permit application form provided by the Zoning Administrator and must include any additional information determined by the Zoning Administrator as necessary to demonstrate compliance with all applicable codes and requirements, along with the ASECS permit application fee, as established by resolution of the Board of Supervisors.
- B. Utility Scale Wind Energy Conversion Systems, Utility Scale Solar Energy Conversion System, and Utility Scale Battery Energy Storage Systems.
 - 1. Renewable Energy Overlay Zoning District Application Requirements. Application for approval of a permit to construct an USWECS, USSECS, or USBESS shall be submitted to the Zoning Administrator on a permit application form provided by the Zoning Administrator and must include any additional information determined by the Zoning Administrator as necessary to demonstrate compliance with all applicable codes and requirements, along with the application fee, as established by resolution of the Board of Supervisors.
 - 2. Permit Procedures. All USWECS, USSECS, or USBESS permit applications shall require review and approval by the Dubuque County Board of Supervisors following the standards and procedures for Renewable Energy Overlay Zoning District. No USWECS, USSECS, or USBESS Building/Zoning Certificate may be issued until after receiving approval of Renewable Energy Overlay Zoning District from the Board of Supervisors and the securing of all other permits as may be required by this chapter and elsewhere in the Dubuque County Ordinances.
- C. Construction Permits. Upon approval of a USWECS, USSECS, or USBESS Permit the following additional permits must be obtained from ~~the~~ Dubuque County ~~Zoning Office~~:
 - 1. Building / Zoning Certificate(s)
 - 2. Entrance permit(s)
 - 3. Address application(s)
 - 4. The following documents are required, if applicable

- a. Site Plan
 - b. Permit for work within Dubuque County right-of-way
 - c. Oversize/Overweight permit (including a plan and agreement to maintain and repair roads impacted by construction activities and traffic)
 - d. Floodplain Development permit
 - e. Erosion and Sediment Control (ESC) permit
 - f. Emergency Response Plan
 - f.g. Plan addressing Project-based Barriers to Animal Movement
- D. Any consulting or legal fees deemed necessary by Dubuque County ~~needed~~ to review the above Utility Scale-USWECS, USSECS, and USBESS applications shall be paid by the applicant in a manner determined and arranged by the Zoning Administrator. Construction Permit fees are as follows:
- a. Accessory WECS - \$300
 - b. Utility Scale-USWECS - \$1,500 for each tower.
 - c. Utility Scale-USSECS and USBESS - \$2 for every \$1,000 of the estimated Project cost, minimum fee of ~~\$3200~~.

5-6 through 5-9 Reserved

**PART 3
DESIGN STANDARDS**

5-10 GENERAL REQUIREMENTS

5-10.1 Wind Energy Conversion Systems (WECS) Design Standards.

- A. Major site plan and rezoning required. A major site plan shall be submitted and reviewed prior to the approval of a USWECS wind farm. The area proposed for the utility-scale-USWECS shall require rezoning to the Renewable Energy Overlay District.
- B. Setbacks.
 - 1. Accessory WECS (AWECS).
 - a. AWECS shall be set back a minimum distance from the base of the structure to all property lines equal to 1.5

- times the height of the tower and rotor as measured from the base to the highest reach of its blade.
- b. AWECS including anchors shall not be located within a required principal structure setback in any zoning district.
- c. An AWECS shall not be located in front of any residential building located on the same parcel.
- 2. Utility Scale WECS (USWECS)
 - a. USWECS shall be a minimum 1,320-feet or 115% of the tower and rotor height whichever is greater from any property lines ~~and road right-of-way lines or residential dwellings~~ not included in the USWECS application, or any public park and/or recreation property line with the following exceptions:
 - i. Any public park or recreational land when approved by the appropriate County, State, or Federal administrative staff, boards, and/or commissions for a demonstrated public purpose.
- C. Rotor Size.
 - 1. AWECS on a parcel with residential as its principal use shall not have a blade diameter in excess of 25-feet.
 - 2. AWECS on a parcel with a nonresidential principal use shall not exceed a 50- foot blade diameter.
- D. Tower Height
 - 1. AWECS shall meet the following requirements:
 - a. AWECS on an individual parcel up to 3-acres shall not exceed a combined tower/pole and rotor height of 65-feet.
 - b. AWECS on a parcel greater than 3-acres and up to 7-acres shall not exceed a combined tower/pole and rotor height of 80-feet.
 - c. AWECS on a parcel greater than 7-acres shall not exceed a combined tower/pole and rotor height of 100-feet.
 - 2. ~~Utility Scale~~ USWECS towers, poles and rotors may exceed the height limitations of the Zoning District in which they are located.
- E. Blade clearance. No portion of a horizontal axis WECS blade shall extend within 30- feet off the ground. No portion of a vertical axis WECS blade shall extend within 20-feet of the ground. No blades may extend over parking areas, driveways, or sidewalks. No blade may extend within 20-feet of the nearest tree, structure, or above ground utility facilities.
- F. Building mounted WECS prohibited. WECS mounted on a roof or wall or

otherwise attached to a building are prohibited.

- G. Tower. Only monopole towers shall be permitted for freestanding WECS. Guy-wire supported mast, lattice, and towers of any other type shall not be considered in compliance with this chapter.
- H. Signage. All signs, both temporary and permanent, are prohibited on WECS, except as follows:
 - 1. Manufacturer's identification on the wind turbine cowling.
 - 2. Appropriate warning signs and placards including visible warning sign of "High Voltage" placed at the base of all conversion systems. The sign shall have at a minimum 6-inch letters with ¾-inch stroke.
 - 3. Identifying name or logo of the owner/operator on the base of some project turbines may be approved by the zoning administrator.
- I. Color. The color of WECS shall be non-reflective and non-obtrusive, preferably white or gray.
- J. Shadow flicker. No WECS shall be installed and operated so to cause a shadow flicker to fall on or in any existing residential dwelling or Iowa DNR regulated Animal Feeding Operation (AFO) including confinement feeding operations or open feedlots, for more than 30 hours per year and no more than 30 minutes per day that is not included as part of the WECS application.
- K. Rotor design and overspeed controls. All WECS shall be equipped with manual and automatic overspeed controls to limit the rotation of blades to a speed below the designed limits. A professional engineer shall certify that the rotor and overspeed control design and fabrication conform to good engineering practices. No changes or alterations from the certified design shall be permitted unless accompanied by a professional engineer's statement of certification.
- L. Electrical compliance. All electrical compartments, storage facilities, wire conduit and interconnections with utility companies shall conform to State electrical codes.
- M. Experimental or prototype AWECS. Written evidence identifying the proposed use of an experimental or prototype WECS shall be submitted to the Zoning Administrator County by a professional engineer and/or factory representative. Experimental or prototype AWECS are not permitted closer than 300-feet from all property lines.
- N. Tower, Poles, and Anchor points. All towers, poles, and anchor points must be unclimbable by design or protected by anti-climbing devices such as:
 - 1. Fences with locking portals at least 6-feet high.
 - 2. Anti-climbing devices 12-feet from base of pole

- O. Noise Levels. Audible noise due to USWECS sites operations shall not exceed fifty (50) dBA for any period of time, when measured at any residential dwelling, school, hospital, church, public library, or Iowa DNR regulated Animal Feeding Operation (AFO) including confinement feeding operations or open feedlots, existing on the date of approval of any Project permit.
- a) In the event audible noise due to USWECS operations contains a steady pure tone, such as a whine, screech, or hum, the standards for audible noise set forth in this subsection shall be reduced by five (5) dBA.
 - b) In the event the ambient noise level (exclusive of the USWECS development in question) exceeds the applicable standard given above, the applicable standard shall be adjusted so as to equal the ambient noise level. The ambient noise level shall be expressed in terms of the highest whole number sound pressure level in dBA, which is succeeded for more than five (5) minutes per hour. Ambient noise levels shall be measured at the exterior of potentially affected existing residences, schools, hospitals, churches, ~~and~~ public libraries, and Iowa DNR regulated animal feeding operations. Ambient noise level measurement techniques shall employ all practical means of reducing the effect of Project generated noise at the measuring microphone. Ambient noise level measurements may be performed when wind velocities at the proposed project site are sufficient to allow wind turbine operation, provided that the wind velocity does not exceed thirty (30) mph at the ambient noise measurement location.
 - c) In the event the noise levels resulting from the USWECS exceed the criteria listed above, a waiver to said levels may be granted by the Board of Supervisors provided that the following has been accomplished:
 - a. Written consent from the affected property owners has been obtained stating that they are aware of the USG-WECS and the noise limitations imposed by this Ordinance, and that consent is granted to allow noise levels to exceed the maximum limits otherwise allowed; and
 - b. If the applicant wishes the waiver to apply to succeeding owners of the property, a permanent noise impact easement shall be recorded in the Office of the Dubuque County Recorder which describes the burdened properties and which advises all subsequent owners of the burdened property that

noise levels in excess of those permitted by this Ordinance may exist on or at the burdened property.

- P. Lighting. Lighting of towers is only allowed when required by the FAA. The lighting method allowed shall be an FAA approved dual lighting system.
- Q. Stormwater Management. Depending on the scale and footprint of a proposed USWECS project, a stormwater management prevention plan may be required in accordance with the site plan regulations of Dubuque County. For the purposes of pollutant removal, stormwater rate and runoff management, flood reduction and associated impacts, the applicant must meet and be in compliance with the Dubuque County Erosion and Sediment Control and Stormwater Ordinance. All site work shall further comply with the National Pollution Discharge Elimination System (NPDES) permit as required by the Iowa Department of Natural Resources (IDNR), including Section 404 of the Clean Water Act (CWA) related to impacts on wetlands and Waters of the United States (WUS).
- R. Emergency Response Plan. A plan for local emergency services on the procedures to respond to a fire or other event on site, the process of clean-up and recovery, and the development of regular training to be provided by the owner/operator of the Project.
 - a. Paved or gravel surfaced road access for emergency service equipment shall be provided and maintained to all structures and buildings.
 - b. The Project shall coordinate with emergency services staff to provide materials, education and/or training to the departments serving the property with emergency services.
 - c. The Project shall provide the servicing Fire Department and all pertinent information in case of a fire or other emergency on site to ensure safe operation and emergency response, including but not limited to specific training, 24-hour contact information, access to lock boxes, access point locations, the locations of shut offs and circulation patterns. The Project shall update the fire departments of any changes to such information and provide an annual tour for any new fire department volunteers. The Project shall provide, at the Project' sole cost, any equipment, keys, or codes as necessary for the fire departments to gain entry to the site in the event of an emergency.
- S. Drainage. USWECS shall follow these drainage requirements:
 - a. The Project shall be responsible for reasonably prompt repair of damage to public drainage systems stemming from construction, operation, or maintenance. In the event of impacts to public

drainage systems within the Project footprint, the Project shall ensure that this does not affect neighboring properties. After notification by the County of any drainage issue, if the applicant or operator fails to take or make a prompt effort to address the drainage issue, the County may act to correct the issue at the applicant's expense.

- b. If drainage infrastructure, such as field tile, or systems on nonparticipating properties are damaged by any cause connected with the Project, the Project shall restore the drainage infrastructure or system to a condition at least as good as the pre-construction condition.

T. Road Use and Mitigation of Damages Plan.

- a. Roads. Applicants shall identify all roads to be used for the purpose of transporting ~~Utility Scale-US~~WECS, substation parts, cement, and/or equipment for construction, operation or maintenance of the ~~US~~WECS and obtain applicable weight and size permits from the impacted road authority(ies) prior to construction.
- b. Existing Road Conditions. Applicant shall conduct a pre-construction survey, in coordination with the impacted local road authority(ies) to determine existing road conditions. The survey shall include photographs and a written agreement to document the condition of the public facility. The applicant is responsible for on-going road maintenance and dust control measures identified by the Dubuque County Engineer during all phases of construction.
- c. Required Financial Security. The applicant shall be responsible for restoring or paying damages as agreed to by the applicable road authority(ies) sufficient to restore the road(s) and bridges to preconstruction conditions. Financial security in a manner approved by the Dubuque County Attorney's Office shall be submitted covering 130% the costs of all required improvements. This requirement may be waived by the Board of Supervisors by recommendation from the County Engineer.

5-10.2 Solar Energy Conversion Systems (SECS) Design Standards.

- A. Accessory Solar Energy Conversion Systems Design Standards. All ASECS shall meet the following standards and conditions:
 - 1. Setback. Solar energy conversion systems must meet the accessory structure setback for the zoning district and principal land use associated with the lot on which the system is located, except as allowed below.

- a. Roof-mounted solar energy conversion systems. The collector surface and mounting devices for roof-mounted solar energy conversion systems shall not extend beyond the exterior perimeter of the building on which the system is mounted or built, unless the collector and mounting system has been explicitly engineered to safely extend beyond the edge, and setback standards are not violated. Exterior piping for solar hot water systems shall be allowed to extend beyond the perimeter of the building on a side-yard exposure. Solar collectors mounted on the sides of buildings and serving as awnings are considered building-integrated systems and are regulated as awnings.
 - b. Ground-mounted solar energy conversion systems. Ground-mounted solar energy conversion systems shall comply with the setback and placement standards in place for accessory structures for the zoning district in which they are located.
- 2. Height. ASECS must meet the following height requirements:
 - a. Building-mounted or roof-mounted solar energy conversion systems shall not exceed the maximum allowed height in any zoning district. For purposes for height measurement, solar energy conversion systems other than building-integrated systems shall be given an equivalent exception to height standards as building-mounted mechanical devices or equipment.
 - b. Ground-mounted or pole-mounted solar energy conversion systems shall not exceed 15-feet in height when oriented at maximum tilt.
 - c. Solar carports in non-residential districts shall not exceed 20-feet in height.
- 3. Visibility. Solar energy conversion systems shall be designed to minimize visual impacts from the public street right-of-way.
 - a. Building Integrated Photovoltaic Systems. Building integrated photovoltaic solar energy conversion systems shall be allowed regardless of whether the system is visible from the public right-of- way, provided the building component in which the system is integrated meets all required setback, land use or performance standards for the district in which the building is located.
 - b. Aesthetic restrictions. Roof-mounted, building integrated, or ground- mounted solar energy conversion systems

should not be restricted for aesthetic reasons if the system is not visible from the closest edge of any public right-of-way other than an alley, or if the system meets the following standards.

- i. Roof-mounted systems on pitched roofs that are visible from the nearest edge of the front right-of-way shall have the same finished pitch as the roof and be no more than ten inches above the roof.
 - ii. Roof-mount systems on flat roofs that are visible from the nearest edge of the front right-of-way shall not be more than 5feet above the finished roof and are exempt from any rooftop equipment or mechanical system screening.
 - c. Glare Minimization. All solar energy conversion systems using a reflector to enhance solar production shall minimize glare from the reflector affecting adjacent or nearby properties. All solar energy conversion systems must not interfere with traffic, including air traffic, or create a safety hazard.
 4. Noise Levels. The noise level measured at the property line of the property on which the SECS has been installed shall not exceed 50 decibels or cause a noise disturbance. In no event shall the SECS create a nuisance.
- B. Utility Scale Solar Energy Conversion Systems (USSECS) Design Standards.
1. Height. USSECS panels or collectors and related solar mounting devices shall not exceed 20-feet in height when panel or collector is oriented at maximum tilt. Projects designed for agri-voltaics may be considered for a higher height restriction.
 2. Setbacks. USSECS, including all accessory structures and equipment, shall be setback a minimum of ~~1,320~~50-feet from all property lines and road right-of-way lines and ~~1,320~~300-feet from any residential dwelling-Iowa DNR regulated Animal Feeding Operation (AFO) including confinement feeding operations or open feedlots, that is not on a participating lot. No setback is required between participating lots.
 3. Visibility. ~~Solar energy conversion systems (USSECS)~~ shall be designed to minimize visual impacts from the public street right-of-way and adjoining property. All solar energy conversion systems using a reflector to enhance solar production shall minimize glare from the reflector affecting adjacent or nearby properties, except for those that are participating lots.

4. Roof-Mounted USSECS. Roof-mounted USSECS shall meet the design standards as described for ASECS in section 5-10.2A.
5. Landscape Buffering. All systems shall comply with the zoning buffers for the zoning district in which located. At the discretion of the Board of Supervisors, additional landscape buffers may be required along any of the project boundaries of the USSECS to buffer adjoining residential land use and public views.
6. Ground Cover and Landscaping. All areas that are not hard or gravel surfaced shall be planted and maintained with a perennial ground cover. Ground around and under solar arrays and in project site buffer areas shall be planted and maintained in perennial vegetated ground cover, and meet the following standards:
 - a. Topsoil shall not be permanently removed during development unless part of a remediation effort.
 - b. Soils shall be planted and maintained in short stature perennial vegetation capable of enhancing soil health and supporting pollinators. Blooming shrubs may be used in buffer areas as appropriate for visual screening.
 - c. Seed mixes and maintenance practices should be consistent with recommendations made by qualified natural resource professionals such as those from the Department of Natural Resources, County Soil and Water Conservation Service or Natural Resource Conservation Service.
 - d. Landowner, farmer, tenant and adjoining landowner notification must be conducted per the terms of an easement or land use agreement, or as otherwise required by the specific Candidate Conservation Agreement or Habitat Conservation Plan under the Endangered Species Act.
7. Stormwater Management. A stormwater management prevention plan may be required in accordance with the site plan regulations of Dubuque County. For the purposes of pollutant removal, stormwater rate and runoff management, flood reduction and associated impacts, the applicant must meet and be in compliance with the Dubuque County Erosion and Sediment Control and Stormwater Ordinance. All site work shall further comply with the National Pollution Discharge Elimination System (NPDES) permit as required by the Iowa Department of Natural Resources (IDNR), including Section 404 of the Clean Water Act (CWA) related to

impacts on wetlands and Waters of the United States (WUS).

8. Lot Coverage. The total amount of impervious surfacing shall not exceed 20% of the site area. For the purpose of this chapter, impervious surfacing includes paved or gravel roads, parking areas, equipment pads, building footprints, and other similar non-vegetated areas. Ground-mounted panel or collector systems shall not be included in this impervious surface calculation if the soil under the panel or collector is maintained in vegetation and not compacted.
9. Emergency Response Plan . A plan for local emergency services on the procedures to respond to a fire or other event on site, the process of clean-up and recovery, and the development of regular training to be provided by the owner/operator of the Project.
 - a. Paved or gravel surfaced road access for emergency service equipment shall be provided and maintained to all structures and buildings.
 - b. The Project shall coordinate with emergency services staff to provide materials, education and/or training to the departments serving the property with emergency services.
 - c. The Project shall provide the servicing Fire Department and all pertinent information in case of a fire or other emergency on site to ensure safe operation and emergency response, including but not limited to specific training, 24-hour contact information, access to lock boxes, access point locations, the locations of shut offs and circulation patterns. The Project shall update the fire departments of any changes to such information and provide an annual tour for any new fire department volunteers. The Project shall provide, at the Project' sole cost, any equipment, keys, or codes as necessary for the fire departments to gain entry to the site in the event of an emergency.
10. All solar energy conversion system components shall be accessible for required routine maintenance without trespassing on adjoining property or disassembling any major portion of the structure or building.
11. Drainage. All ~~Utility Scale~~ USSECS shall follow these drainage requirements:
 - a. The Project shall be responsible for reasonably prompt repair of damage to public drainage systems stemming from construction, operation, or maintenance. In the event

of impacts to public drainage systems within the Project footprint, the Project shall ensure that this does not affect neighboring properties. After notification by the County of any drainage issue, if the applicant or operator fails to take or make a prompt effort to address the drainage issue, the County may act to correct the issue at the applicant's expense.

- b. If drainage infrastructure, such as field tile, or systems on nonparticipating properties are damaged by any cause connected with the Project, the Project shall restore the drainage infrastructure or system to a condition at least as good as the pre-construction condition.

12. Road Use and Mitigation of Damages Plan.

- a. Roads. Applicants shall identify all roads to be used for the purpose of transporting ~~Utility Scale~~ USSECS, substation parts, cement, and/or equipment for construction, operation or maintenance of the USSECS and obtain applicable weight and size permits from the impacted road authority(ies) prior to construction.
- b. Existing Road Conditions. Applicant shall conduct a pre-construction survey, in coordination with the impacted local road authority(ies) to determine existing road conditions. The survey shall include photographs and a written agreement to document the condition of the public facility. The applicant is responsible for on-going road maintenance and dust control measures identified by the Dubuque County Engineer during all phases of construction.
- c. Required Financial Security. The applicant shall be responsible for restoring or paying damages as agreed to by the applicable road authority(ies) sufficient to restore the road(s) and bridges to preconstruction conditions. Financial security in a manner approved by the Dubuque County Attorney's Office shall be submitted covering 130% the costs of all required improvements. This requirement may be waived by the Board of Supervisors by recommendation from the County Engineer.

5-10.3 Battery Energy Storage Systems (BESS) Design Standards

- A. Height. ~~Utility Scale~~ US-BESS equipment, buildings, and related

structures shall not exceed 50-feet in height.

- B. Setbacks. ~~Utility Scale~~US-BESS equipment, including all accessory structures, shall be setback a minimum of ~~1,320~~100-feet from all property lines and road right-of-way lines. No setback is required between participating lots.
- C. Landscape Buffering. All ~~Utility Scale~~US-BESS equipment shall comply with the zoning buffers for the zoning district in which it is located. At the discretion of the Board of Supervisors, a landscape buffer may be required along any of the project boundaries of the USBESS to buffer adjoining residential land use and public views.
- D. Ground Cover and Landscaping. All area not hard surfaced or graveled shall be planted and maintained with a perennial ground cover.
- E. Stormwater Management. A stormwater management prevention plan may be required in accordance with the site plan regulations of Dubuque County. For the purposes of pollutant removal, stormwater rate and runoff management, flood reduction and associated impacts, the applicant must meet and be in compliance with the Dubuque County Erosion and Sediment Control and Stormwater Ordinance.
- F. Lot Coverage. The total amount of impervious surfacing shall not exceed 25% of the site area. For the purpose of this chapter, impervious surfacing includes paved or gravel roads, parking areas, equipment pads, building footprints, and other similar non-vegetated areas.
- G. Emergency Access. Paved access for emergency service equipment shall be provided and maintained to all ~~Utility Scale~~US-BESS equipment and related structures.
- H. Emergency Response Plan. A plan for local emergency services on the procedures to respond to a fire or other event on site, the process of clean-up and recovery, and the development of regular training to be provided by the owner/operator of the facility.
 - a. Paved or gravel surfaced road access for emergency service equipment shall be provided and maintained to all structures and buildings.
 - b. The Project shall coordinate with emergency services staff to provide materials, education and/or training to the departments serving the property with emergency services.
 - c. The Project shall provide the servicing Fire Department and all pertinent information in case of a fire or other emergency on site to ensure safe operation and emergency

response, including but not limited to specific training, 24-hour contact information, access to lock boxes, access point locations, the locations of shut offs and circulation patterns. The Project shall update the fire departments of any changes to such information and provide an annual tour for any new fire department volunteers. The Project shall provide, at the Project' sole cost, any equipment, keys, or codes as necessary for the fire departments to gain entry to the site in the event of an emergency.

- I. Noise Levels. Audible noise due to ~~Utility Scale~~US-BESS sites operations shall not exceed fifty (50) dBA for any period of time, when measured at any residential dwelling, school, hospital, church, public library, or Iowa DNR regulated Animal Feeding Operation (AFO) including confinement feeding operations or open feedlots existing on the date of approval of any Project permit.
 - a. In the event audible noise due to USBESS operations contains a steady pure tone, such as a whine, screech, or hum, the standards for audible noise set forth in this subsection shall be reduced by five (5) dBA.
 - b. In the event the ambient noise level (exclusive of the USBESS development in question) exceeds the applicable standard given above, the applicable standard shall be adjusted so as to equal the ambient noise level. The ambient noise level shall be expressed in terms of the highest whole number sound pressure level in dBA, which is succeeded for more than five (5) minutes per hour. Ambient noise levels shall be measured at the exterior of potentially affected existing residences, schools, hospitals, churches, ~~and~~ public libraries, or Iowa DNR regulated Animal Feeding Operation(AFO). Ambient noise level measurement techniques shall employ all practical means of reducing the effect of Project generated noise at the measuring microphone. Ambient noise level measurements may be performed when wind velocities at the proposed project site are sufficient to allow wind turbine operation, provided that the wind velocity does not exceed thirty (30) mph at the ambient noise measurement location.
 - c. In the event the noise levels resulting from the USBESS exceed the criteria listed above, a waiver to said levels may be granted by the Board of Supervisors provided that the following has been accomplished:

- i. Written consent from the affected property owners has been obtained stating that they are aware of the USBESS and the noise limitations imposed by this Ordinance, and that consent is granted to allow noise levels to exceed the maximum limits otherwise allowed; and
- ii. If the applicant wishes the waiver to apply to succeeding owners of the property, a permanent noise impact easement shall be recorded in the Office of the Dubuque County Recorder which describes the burdened properties and which advises all subsequent owners of the burdened property that noise levels in excess of those permitted by this Ordinance may exist on or at the burdened property.

5-11 through 5-19 Reserved

PART 4
DECOMMISSIONING AND RECLAMATION PLAN

5-20 Decommissioning and Reclamation.

5-20.1 Plan Required All USWECS, USSECS, and USBESS permit applications shall include a Decommissioning and Reclamation Plan that includes the following:

- A. The anticipated start and completion dates and manner in which the project will be decommissioned, including a description of which above-grade and below-grade improvements will be removed, retained (e.g., access drive, fencing), or restored for viable reuse of the property, including equipment disposal plan that complies with state code and administrative rules.
- B. The projected decommissioning costs for removal and disposal of the USWECS, USSECS, or USBESS (net of salvage value in current dollars) and restoration of the site and soil for agricultural use, certified by an Iowa licensed engineer.
- C. The method of ensuring that funds will be available for site decommissioning and restoration of the site (e.g., surety bond, cash deposit).

- D. Restoration or reclamation activities outlined in the Plan shall include but not be limited to the following:
- a. Restoration of the pre-construction surface grade and soil profile after removal of structures, equipment, graveled areas and access roads.
 - b. Re-vegetation of restored soil areas with crops, native seed mixes, plant species suitable to the area, consistent with the County's weed control plan.
 - c. For any part of the Project on leased property, the Plan may incorporate agreement with the landowner regarding leaving access roads, fences, gates, or repurposed buildings in place or regarding restoration of agricultural crops or forest resource land. Any use of remaining structures must be in conformance with the zoning regulations that would be in effect after the expiration or termination of the Renewable Energy Overlay Zoning District.

5-20.2 Plan Review and approval The Decommissioning and Reclamation Plan shall be reviewed by the Board of Supervisors as part of the Permit application and included as a condition of approval of any permit. The Board of Supervisors ~~County~~ reserves the right to verify that adequate decommissioning terms are contained in any landowner easements.

5-20.3 Agreement The decommissioning and reclamation plan will be provided in the form of an executed agreement, recorded against the property, and binding upon the owner/operator and any of their successors, assigns, or heirs.

5-20.4 Regular Review Required

- A. Every 5-years from the date of approval of the permit until its expiration, the owner/operator of a USWECS, USSECS, and USBESS approved under this chapter shall cause a review to be completed of the decommissioning and reclamation plan and the amount of the performance guarantee required based on inflation, salvage value, and current removal costs. A written report of this review, certified by an Iowa licensed engineer, shall be provided to the Zoning Administrator within the above timeline.
- B. Based on the findings of the report, the Zoning Administrator may require additional surety be provided or may allow a reduction in the required amount of surety. Failure to provide, within a reasonable time, the additional surety as may be required by the Zoning Administrator shall be considered a violation of the permit approval and said permit may be revoked by the Board of Supervisors ~~Dubuque County~~.

- C. Failure to provide this report within the above timeline shall be considered a violation of the permit approval and said permit may be revoked by ~~the Board of Supervisors Dubuque County~~.
- D. At the request of the owner/operator, an amendment to the decommissioning and reclamation plan agreement may be presented to the Board of Supervisors for their review and consideration.

5-21 through 5-24 Reserved

PART 5

INSTALLATION AND CONSTRUCTION CERTIFICATION

5-25 Installation and Construction Certification

- 5-25.1** All approved USWECS, USSECS, and USBESS projects shall commence installation, which can include substantial progress on project engineering and design, as specified in the USWECS, USSECS, and USBESS Permit, and in accordance with this ordinance, within one year of approval or the Building / Zoning Certificate expires. Expired Certificates require a new application. A new application and fee must be resubmitted.
- 5-25.2** A professional engineer shall certify that the construction and installation of the USWECS, USSECS, or USBESS meets or exceeds the manufacturer's construction and installation standards, prior to the property owner and/or applicant receiving a Permit for the USWECS, USSECS, or USBESS.
- 5-25.3** Construction hours shall be daylight hours, not earlier than 7:00am and not later than 7:00pm, Monday through Saturday. For this purpose, construction does not include meetings of crews, operation of cars, light trucks, or forklifts. Exceptions to this shall be made on a case-by-case basis by the Zoning Administrator.
- 5-25.4** Construction laydown areas shall be located in a manner to minimize impacts to nonparticipating residences and shall be set back at least 500 feet from nonparticipating residences.

5-26 through 5-29 Reserved

PART 6
MAINTENANCE AND OPERATION

5-30 Maintenance and Operation

- 5-30.1** The project Owner and/or Operator is responsible for maintaining, repairing, and keeping USWECS, USSECS, or USBESS in good working order.
- A. Maintenance shall include, but not be limited to, painting, structural repairs, monitoring and mitigation of stray voltage, and integrity of security measures. Site access shall be maintained to a level acceptable to emergency response officials. Any retrofit, replacement or refurbishment of equipment shall adhere to all applicable local, state and federal requirements.
 - B. Waste disposal of solid and/or hazardous materials shall occur in accordance to County, State and Federal requirements. No worn parts, used oils or lubricants, crates or other items shall be located on the USWECS, USSECS, or USBESS property or project area. Construction and other bulk materials and replacement parts and components shall not be stored outside on the USWECS, USSECS, or USBESS property or project area.
 - C. The USWECS, USSECS, or USBESS shall not cause interference with microwave transmissions, residential television interference, or other communication equipment. If issues with radio reception are identified after installation, the owner/operator is responsible to mitigate the issue.
 - D. Access for emergency service equipment shall be maintained.
 - E. The Zoning Administrator/Building Official and any other necessary personnel may enter the property with guided access from the project owner or their representative for which a Permit or Building Permit has been issued under this ordinance to conduct an inspection to determine whether the conditions stated in the permit have been met as specified by statute, ordinance, or code. Failure to provide access shall be deemed a violation of this ordinance.

5-31 through 5-34 Reserved

PART 7
REMOVAL, REVOCATION, AND ABANDONMENT

5-35 Removal, Revocation, and Abandonment

5-35.1 The USWECS, USSECS, or USBESS shall be removed if not operated for greater than 12-months or fails to be operable for 12-consecutive months or if it is determined by Dubuque County that a USWECS, USSECS, or USBESS is not in compliance with provisions of this ordinance and the associated Permits.

- A. As a condition of Permit approval for all USWECS, USSECS, or USBESS the project Owner and/or Operator shall sign a “Consent to Removal” form supplied by the Zoning Administrator. The “Consent to Removal” will state that the owner shall remove the USWECS, USSECS, or USBESS or allow the Board of Supervisors Dubuque County to remove and assess the property for the cost to remove the USWECS, USSECS, or USBESS. Upon noncompliance the County shall send a notice to the property owner and/or USWECS, USSECS, or USBESS owner by certified mail identifying the options to correct the violations. If the corrective actions are not taken within 30- days of certified mailing date a final notice will be sent to the owner in the same manner and be given 30-additional days to properly correct the violations. If the violations are not corrected, the USWECS, USSECS, or USBESS will be removed by the Board of Supervisors Dubuque County and the property will be assessed for the cost of removal and disposal.
- B. As a condition of approval of any USWECS, USSECS, and USBESS permit, the project Owner and/or Operator shall provide sufficient surety to cover the cost of the removal and disposal of said USWECS, USSECS, and USBESS facility. The final amount and form of the surety shall be determined by the ~~Zoning Administrator~~Board of Supervisors.

5-35.2 A USWECS, USSECS, or USBESS Permit may be revoked any time the Project does not comply with the rules and regulations set forth in this ordinance or Permit. The revocation of the Permit requires the USWECS, USSECS, or USBESS to be physically removed within 12 months.~~ninety (90) days~~.

5-35.3 At such time that a USWECS, USSECS, or USBESS is abandoned or discontinued, the property owner and or Project owner will notify the Dubuque County Zoning Administrator by certified U.S. mail of the proposed date of abandonment or discontinuation of operations and initiate removal of the Project within 12 months.

5-35.4 Upon revocation, abandonment, or discontinuation of USWECS, USSECS, or USBESS or failure to operate, maintain and keep in good working order, the

project Owner and/or Operator shall physically remove the USWECS, USSECS, or USBESS according to the provisions of this ordinance, conditions added to the Permit and/or the “Consent to Removal”. “Physically remove” shall include, but not be limited to:

- A. Removal of the USWECS, USSECS, or USBESS systems 4-feet below ground and above, including any turbines, towers, poles, panels, collectors, mounting devices, electrical equipment, access roads and drives, parking areas, equipment pads, buildings, structures, above ground foundations, and overhead utility lines related to the operation of the Project.
- B. Restoration of the location of the USWECS, USSECS, or USBESS to its natural condition, except that any landscaping or grading, may remain.

5-35.5 At such time that a USWECS, USSECS, or USBESS Permit is revoked, or is abandoned, discontinued, and required notification has been provided to the ~~County~~ Zoning Administrator, and before required system removal has occurred, a Reactivation Plan may be submitted for the Project, or portions of it, to the Zoning Administrator outlining the steps, costs, and schedule for returning the USWECS, USSECS, or USBESS to operational service. Approval of the reactivation plan is determined by the Board of Supervisors.

5-36 through 5-39 Reserved

**PART 8
RELEASE OF LIABILITY**

5-40 Release of Liability

5-40.1 Dubuque County shall be fully released of any liability associated with any WECS, SECS, or BESS built in unincorporated Dubuque County.

5-41 through 5-44 Reserved

**PART 9
WIND AND SOLAR ACCESS EASEMENTS**

5-45 Wind and Solar Access Easements

- 5-45.1** The grant of any permit under this chapter does not constitute the granting of an easement by Dubuque County. The WECS owner/operator shall have the sole responsibility to acquire any covenants, easements, or similar documentation to assure and/or protect access to sufficient wind as may or may not be necessary to operate the WECS. Similarly, the owner/operator of any SECS shall have the sole responsibility to acquire any covenants, easements, or similar documentation to assure and/or protect access to sufficient sunlight as may or may not be necessary to operate the system.

5-46 through 5-49 Reserved

**PART 10
SEVERABILITY**

- 5-50** **SEVERABILITY.** If any section, provision, or part of this ordinance shall be adjudged invalid or unconstitutional, such adjudication shall not affect the validity of the regulations as a whole or any section, provision, or part thereof not adjudged invalid or unconstitutional.

5-51 through 5-54 Reserved

**PART 11
EFFECTIVE DATE**

- 5-55** **EFFECTIVE DATE.** This ordinance shall take effect upon its publication as required by law.

5-56 through 5-59 Reserved

**PART 12
CONFLICT WITH PROVISIONS**

- 5-60** **CONFLICT WITH PROVISIONS.** All ordinances or parts of ordinances in conflict with the provisions of this ordinance are hereby repealed.